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Inorganic Cumulative Examination
Fall 1999

Electronic spin is an important consideration in the chemistry of transition metals. Rationalize, in detail, the following observations in terms of the electronic properties of the complexes.

1. In solution, many complexes of Fe(II) that contain simple ligands such as Cl^-, water and ammonia have high magnetic moments. Other complexes such as those containing phenanthroline ligands are diamagnetic. Combinations of these ligands produce complexes that have strongly temperature dependent magnetic moments. The moments can be near zero at low temperatures and rise quickly at high temperatures.

2. The intensities of the absorption bands of many transition metals are very weak with extinction coefficients on the order of 1-100. Simple aqueous solutions of manganese(II) show bands that are particularly weak with extinction coefficients very much less than 1.

3. Most of the transition metals complexes have excited states with lifetimes on the pico or femtosecond time scale. A few complexes have absorption bands with very large extinction coefficients and excited-state lifetimes on the order of nanoseconds.